

NoSQL-Mongo Lab3

Notes

- If a **Key** has no value → add any value.
- If test data is missing → insert it before testing.
- **Verify and test** your data, and provide a **justification** for your answer.

Use Robo 3 T to create the following:

1. Provide the MongoDB code for enforcing JSON schema validation when creating a collection named "employees" with required fields "name," "age" (min. 18), and "department" (limited to ["HR," "Engineering," "Finance"]).
2. Create new Database named **Demo**

And Collections named **trainingCenter1**, **trainingCenter2**

- a. Insert documents into **trainingCenter1** collection contains (Use Variable named data as Array)
 - i. `_id`, name as firstName lastName , age , address as array , status
 - b. Using insert ONE from **data** Variable
 - c. Using Same Variable (data) with same data and insert MANY into **trainingCenter2** collection
3. Use find. explain function (find by age field) and *mention* scanning type
 4. Create index on created collection named it “IX_age” on age field
 5. Use find. explain view winning plan for index created (find by age field) and *mention* scanning type
 6. Create index on created collection named it “compound” on firstName and lastName
 - a. Try find().explain before create index and *mention* scanning type
 - b. Try find().explain after create index and *mention* scanning type
 7. Drop **Demo** Database

Bonus Part

1. Use **mongodump** to back up your Lab database.
2. **Drop** the Lab database.
3. Use **mongorestore** to restore it with a new name: **ITI_Course**.